Question 1

Display all environment variables using the more command

Approach

I listed all environment variables and piped the output into the more command to view them page by page.

Commands

- `env` → shows environment variables
- `printenv` → alternative command
- `more` → displays output page by page

Output

STEP 1 : view environment variables

```
┌──(aman㉿kali)-[~]
└─$ env | more
COLORFGBG=15;0
COLORTERM=truecolor
COMMAND_NOT_FOUND_INSTALL_PROMPT=1
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
DESKTOP_SESSION=lightdm-xsession
DISPLAY=:0.0
DOTNET_CLI_TELEMETRY_OPTOUT=1
GDMSESSION=lightdm-xsession
HOME=/home/aman
LANG=en_IN
LANGUAGE=en_IN:en
LOGNAME=aman
NMAP_PRIVILEGED=
PANEL_GDK_CORE_DEVICE_EVENTS=0
PATH=/home/aman/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/aman/.dotnet/tools
POWERSHELL_TELEMETRY_OPTOUT=1
POWERSHELL_UPDATECHECK=Off
PWD=/home/aman
QT_ACCESSIBILITY=1
QT_AUTO_SCREEN_SCALE_FACTOR=0
QT_QPA_PLATFORMTHEME=qt5ct
SESSION_MANAGER=local/kali:@/tmp/.ICE-unix/2117,unix/kali:/tmp/.ICE-unix/2117
SHELL=/usr/bin/zsh
SHLVL=1
SSH_AGENT_PID=2268
SSH_AUTH_SOCK=/home/aman/.ssh/agent/s.RDJpj98jBr.agent.U13EgfJunK
TERM=xterm-256color
USER=aman
WINDOWID=0
XAUTHORITY=/home/aman/.Xauthority
XDG_CACHE_HOME=/home/aman/.cache
XDG_CONFIG_DIRS=/etc/xdg
XDG_CONFIG_HOME=/home/aman/.config
XDG_CURRENT_DESKTOP=XFCE
XDG_DATA_DIRS=/usr/share/xfce4:/usr/local/share/:/usr/share/:/usr/share
XDG_GREETER_DATA_DIR=/var/lib/lightdm/data/aman
XDG_MENU_PREFIX=xfce-
XDG_RUNTIME_DIR=/run/user/1000
XDG_SEAT=seat0
XDG_SEAT_PATH=/org/freedesktop/DisplayManager/Seat0
XDG_SESSION_CLASS=user
XDG_SESSION_DESKTOP=lightdm-xsession
XDG_SESSION_ID=60
XDG_SESSION_PATH=/org/freedesktop/DisplayManager/Session0
XDG_SESSION_TYPE=x11
XDG_VTNR=7
OLDPWD=/home/aman
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.7z=01;31:*.ace=01;31:*.alz=01;31:*.apk=01;31:*.arc=01;31:*.arj=01
;31:*.bz=01;31:*.bz2=01;31:*.cab=01;31:*.cpio=01;31:*.crate=01;31:*.deb=01;31:*.drpm=01;31:*.dwm=01;31:*.dz=01;31:*.ear=01;31:*.egg=01;31:*.esd=01;31:*.gz=01;31:*.jar=01;31:*.lha=01;31:*.lrz=01;31:*.lz=01;31:*.lz4=01;31:*.lzh=01;31:*.l
zma=01;31:*.lzo=01;31:*.pyz=01;31:*.rar=01;31:*.rpm=01;31:*.rz=01;31:*.sar=01;31:*.swm=01;31:*.t7z=01;31:*.tar=01;31:*.taz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tgz=01;31:*.tlz=01;31:*.txz=01;31:*.tz=01;31:*.tzo=01;31:*.tzst=01;31:*.udeb=01
```

Key Observations

Environment variables store system configuration values such as :

- USER
- HOME
- PATH

- SHELL

What I learned

Environment variables store important system configuration information.

Question 2

Display the HOSTNAME variable using echo

Approach

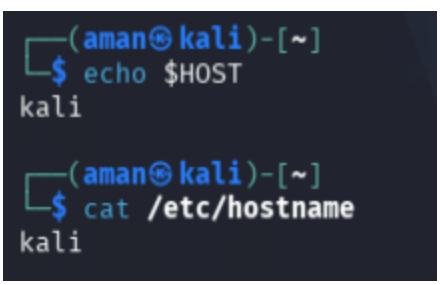
I used the echo command to print the value stored in the HOSTNAME variable.

Commands

- `echo` → prints variable value
- `$HOSTNAME` → stores system hostname

Output

STEP 1 : display hostname variable



Key Observations

HOSTNAME identifies the system name on the network.

What I learned

Environment variables can be accessed using the `$` symbol.

Question 3

Modify the PS1 variable to change forward slash to backslash

Approach

I modified the PS1 environment variable to change the shell prompt appearance.

Commands

- `PS1` → controls terminal prompt appearance.

Output

STEP 1 : modify PS1 prompt

```
┌──(aman㉿kali)-[~]
└─$ PS1='C:\\>'

C:\>
```

Key Observations

Changing PS1 alters how the terminal prompt is displayed.

What I learned

PS1 variable customizes command line appearance.

Question 4

Create a new variable named MYNEWVARIABLE and assign a value

Approach

I created a custom environment variable and assigned my name as its value.

Commands

- variable format : `VARIABLE_NAME=value`

Output

STEP 1 : create variable

```
┌──(aman㉿kali)-[~]
└─$ MYNEWVARIABLE=man
```

Key Observations

Variables are created without spaces around the equals sign.

What I learned

Custom variables can store user-defined values.

Question 5

Display the value stored in MYNEWVARIABLE

Approach

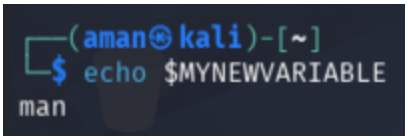
I used echo command to display the stored value.

Commands

- `$VARIABLE_NAME` → retrieves stored value

Output

STEP 1 : display variable value



Key Observations

Echo prints the stored variable value.

What I learned

Variables store reusable values accessible throughout the session.

Question 6

Export MYNEWVARIABLE so it is available globally

Approach

I used the export command to make the variable available across different shell environments.

Commands

- `export` → makes variable available globally

Output

STEP 1 : export variable

```
┌──(aman㉿kali)-[~]
└─$ export MYNEWVARIABLE
```

STEP 2 : verify export

```
┌──(aman㉿kali)-[~]
└─$ printenv MYNEWVARIABLE
man
```

Key Observations

Exported variables can be accessed by child processes.

What I learned

Export allows variables to persist across different shell sessions.

Question 7

Display the PATH variable

Approach

I used echo to display directories stored in the PATH variable.

Commands

- `PATH` → list of executable directories

Output

STEP 1 : display PATH variable

```
┌──(aman㉿kali)-[~]
└─$ echo $PATH
/home/aman/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/aman/.dotnet/tools
```

Key Observations

PATH contains directories separated by colon (:)
`/usr/bin:/bon:/usr/sbin`

What I learned

PATH determines where the system looks for executable files.

Question 8

Add home directory to PATH variable

Approach

I modified PATH variable to include my home directory.

Commands

- `$HOME` → home directory variable

Output

STEP 1 : add home directory to PATH

```
┌──(aman㉿kali)-[~]
└─$ export PATH=$PATH:$HOME
```

STEP 2 : verify PATH update

```
┌──(aman㉿kali)-[~]
└─$ echo $PATH
/home/aman/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/aman/.dotnet/tools:/home/aman
```

Key Observations

New directory is appended to PATH variable.

What I learned

PATH can be modified to allow execution of programs from custom directories.

Question 9

Modify PS1 variable to display custom prompt message

Approach

I customized the PS1 variable to change terminal prompt text.

Commands

- PS1 → controls terminal prompt format

Output

STEP 1 : change prompt text

```
┌──(aman㉿kali)-[~]
└─$ PS1="Wold's Greatest Hacker: "

Wold's Greatest Hacker:
```

Key Observations

Terminal prompt text changes immediately.

What I learned

PS1 customization helps personalize terminal environment.

Security Insights

1. Environment variables may reveal sensitive system configuration details.
2. Hostname information can help attackers identify target systems.
3. PS1 manipulation can disguise malicious terminal environments.
4. Custom variables can store sensitive or malicious values.
5. Viewing variable contents helps identify exposed secrets.
6. Exported variables may affect multiple programs and sessions.
7. PATH manipulation can allow execution of malicious binaries.
8. Attackers may add hidden directories to PATH for persistence.
9. Prompt modification may be used in social engineering attacks.